

When Can Language Models Join Without Lexical Cues?

Unique-Path Traversal versus End-to-End Accuracy under Path Ambiguity*

Anonymous authors
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Abstract

We ask whether multi-hop answering over knowledge graphs needs readable predicates, or only equality among identifiers. **Opaque isomorphic reasoning (OIR)** is a measurement protocol: a renaming σ of entities and relations that we verify to be injective on each instance’s finite atom set, with no legend in the prompt. HMAC-SHA256 is one generator of σ , not a cryptographic or privacy primitive.

Finding. Unique sealed routes still join. Competing sealed routes typically yield UNKNOWN, not the decoy. With the start already in q , OpenAI opaque unique is 32/32; both-opaque unique is 32/32 for Composer 2.5, OpenAI `gpt-5.6-so1`, and Grok 4.5. A city-typed OpenAI freeze at $n=200$ (every HQ is Wikidata P31 city-like; QIDs re-checked live) is unique English/opaque 200/200, English two-path 199/200, named-arm opaque two-path 38/200 (Wilson [0.14, 0.25]; UNKNOWN 151, decoy 11), plan 200/200. The $n=26$ city slice of the $n=32$ lock had decoy 0; that sample cannot resolve a $\sim 5\%$ decoy rate. At $n=200$ the ratio is about 14:1 UNKNOWN to decoy. Exact-match under two sealed routes is strongly sensitive to prompt surface form (Sclar et al., 2024). At $n=200$, same-date 512-token case-id wording (ARM line kept) is named-arm 38/200 vs. hashed-id H5-cyclic 134/200. At 4096/medium the ARM line is non-monotonic and interacts with decoy type (cyclic/constant: no ARM 174/151, H5 129/44, K2 179/144). English relation names restore OpenAI with anonymous entities (OE two-path 32/32 vs. both-opaque 1/32). We score the answer node, not an emitted (r_1, r_2) . Seals remove lexical cues; they are not a confidentiality proof.

1 Introduction

The scored Wikidata question is *What city is the headquarters of the company led by Michael Eisner?* English unique-path context walks `works_at` then `headquartered_in` to Burbank; that success may use English predicates, or only equality among identifiers. The two-path decoy is a second person $\rightarrow$ company $\rightarrow$ city route (`partner_of/located_in`) to Kansas City. Opaque two-path keeps the English question and entity names; those predicates become HMAC tokens (`works_at` $\mapsto$ E334675310f05 on the scored Eisner item). Figure 1 is the $n=32$ OpenAI discovery 2×2 . A city-typed OpenAI $n=200$ freeze (Appendix A) is unique-path 200/200 and named-arm opaque two-path 38/200 (UNKNOWN 151, decoy 11). The $n=26$ city slice of the $n=32$ lock had decoy 0; that zero is not the $n=200$ rate. Prompt-surface packing is Section 5.2.

OIR measures whether a model can join after English predicate strings are replaced by keyed opaque tokens. We write a 2-hop as two inferred maps (only the answer node is scored): selection $q \mapsto (r_1, r_2)$ and traversal $(v_0, r_1, r_2, G) \mapsto v_2$. Opacity does not, by itself, block traversal. The question is when the prompt no longer maps query semantics onto one of two sealed relation sequences. Putting the start entity in q as the same atom as in V (**start in q**) separates that question from start-entity identification.

*Large language models were used as writing assistants for grammar and phrasing. All ideas, claims, experimental design, locked numbers, and scientific conclusions are the authors’.

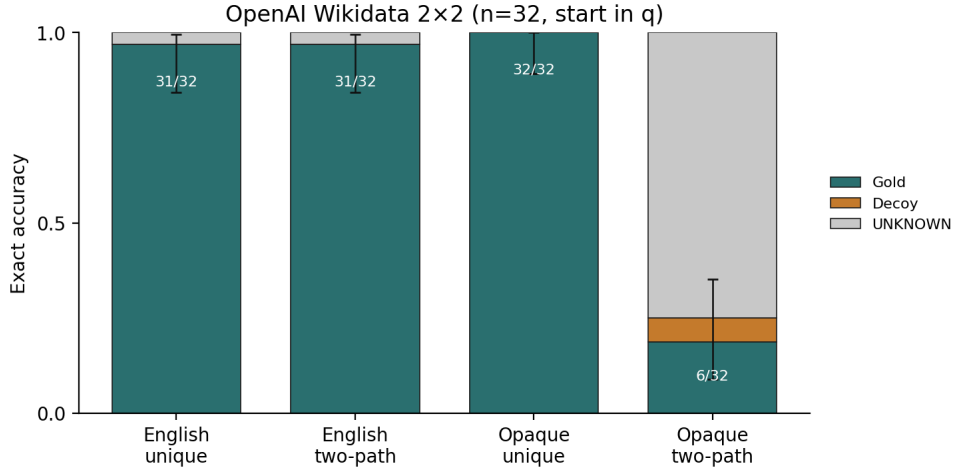


Figure 1: OpenAI WIKI-H5 discovery plot ($n=32$, start in q). Stacked bars are gold / decoy / UNKNOWN; whiskers are Wilson 95% intervals (one August draw). Named-arm opaque two-path 6/32 (mostly UNKNOWN); city slice 5/26 had decoy 0. City-typed $n=200$ (not plotted): 38/200 gold, 151 UNKNOWN, 11 decoy. Prompt-surface ablations are Section 5.2.

Each instance uses a keyed injective σ on that instance’s atom set (HMAC-SHA256, 48-bit tokens; Section 4). The model emits the answer node in the graph’s namespace or UNKNOWN; recovering σ^{-1} is not scored. A written hop is an execution control. Matched versus novel demonstrations test in-prompt reuse of shown seals.

OIR is a measurement protocol. HMAC renaming is the same family as PrivGemo’s entity map ϕ_Q (Tan et al., 2026); we apply it to relation strings, which that paper leaves as original or clustered schema labels. “Lexical cues” on the primary 2×2 means relation strings (entities stay English); both-opaque is a separate ablation. English multi-hop scores should not be read as evidence that the same join succeeds once predicates are sealed and two same-type routes compete.

This paper reports: (i) unique-route sealed execution with the start in q and no written hop; (ii) under two competing sealed routes the usual reply is UNKNOWN, not the decoy, and exact-match moves with prompt surface form (case-id wording; ARM line non-monotonic; Section 5.2); (iii) matched demonstrations restore shown seals, while novel seals do not. What is new is the interaction of opacity with path ambiguity, not opacity alone, and how large a prompt-surface move can be on the same sealed graph. Cited privacy KGQA keeps relation strings as readable words (Ning et al., 2026; Tan et al., 2026; Liu et al., 2026). We seal *relation* strings, keep entities English, and cross that with path ambiguity. Unique routes succeed; two competing same-type routes are underspecified and packaging-sensitive (Table 2).

2 Related Work

Think-on-Graph and Reasoning-on-Graphs select the next readable relation (Sun et al., 2024; Luo et al., 2024). Ciphered ICL is not path ambiguity on a labeled graph (Fang et al., 2025; Guo et al., 2026). GSM-Symbolic swaps entity names in English math (Mirzadeh et al., 2025); 2WikiMultihopQA is a readable multi-hop control (Ho et al., 2020).

ARoG anonymizes *entities* while relations stay the retrieval language (Ning et al., 2026). PrivGemo HMAC-seals entities (ϕ_Q , e.g. **ent_17**); relations stay original labels or clustered schema words, not keyed opaque tokens, and a local Hand answers on the raw graph after a remote Brain ranks retrieved paths (CWQ 67% $\rightarrow$ 59%; WebQSP 75%) (Tan et al., 2026). CoR explores readable relations because entities often have no semantics beyond opaque IDs — the OE hide-set, not relation-only WIKI-H5 (Liu et al., 2026). Tang et al. (2023) find LLMs fail when relation names are replaced by symbols (**r1**) and rules must carry deduction;

Table 1: Related protocols. PrivGemo HMAC-seals entities; relations remain original or clustered schema labels. OIR sends one graph and scores the answer token from the same model.

Work	Hidden	Remote task	Join under opacity
ICL	substitution text	ICL task accuracy	ICL tasks
CIPHERS		(bijection inferred)	
ARoG	entities; relations readable; topic names in q	retrieve + generate (relations, concepts)	privacy KG-RAG
PrivGemo	entities (HMAC); relations original or clustered schema words	rank retrieved paths; local plaintext answer	privacy KG-RAG
CoR	opaque entity IDs; relations readable	relation-centric exploration	KGQA
OIR	relations (primary 2×2); entities too (ablation); one join or none	emit graph-namespace answer (no local decode)	selection vs. traversal

OpenAI unique-path opaque 32/32 is the complementary case: sealed relations still traverse when topology fixes the route. Guo et al. (2025) report collapse to random guessing among end entities when distractor two-hops have other sources. Named-arm Wiki-H5 opaque two-path, with UNKNOWN allowed, is abstention rather than guessing (OpenAI decoy 2/32, UNKNOWN 24/32; at $n=200$, 11 vs. 151). Section 5.2 shows that preferring UNKNOWN is a named-arm fact for at least Grok: no-ARM opaque commits are near chance. A forced-choice arm is not in this evaluation (Section 7) and would put those protocols on the same scale. OIR withholds retrieved paths on the free arms (Table 1). The primary cell seals relations while entities stay English; both-opaque is a separate ablation.

Chain-of-thought and Text-to-SQL externalize plans over English (Wei et al., 2022; Yu et al., 2018). We ask a behavioral question: can a pretrained model join when the relation vocabulary is freshly randomized and no plan is given? A sealed plan is then an execution upper bound. NLGraph and Talk like a Graph encode graphs without readable typed predicates (Wang et al., 2023; Fatemi et al., 2024); OIR starts from a labeled KG and removes only the relation lexicon. ARM headers are a prompt-surface feature (Sclar et al., 2024). Symbol tuning, binding-ID analyses, and compositional-limit studies change the labels or the task (Wei et al., 2023; Feng and Steinhardt, 2024; Dziri et al., 2023). OIR holds the join, hop length, graph size, and question template fixed and replaces relation strings; OpenAI unique-path opaque traversal stays at 32/32, so that removal withdraws the cue that would otherwise select between routes rather than blocking execution.

3 Problem Formulation

Let $G = (V, R, T)$ be a labeled multi-relational graph ($T \subseteq V \times R \times V$) and q a natural-language or programmatic query with answer $a \in V$. A **seal** σ is a keyed map from entity and relation strings to opaque tokens that we verify to be injective on the finite normalized atom set of each evaluation instance. We reject any instance where two distinct raw atoms normalize to the same string. Truncated HMAC-SHA256 is not injective over $\{0, 1\}^*$; we use 48-bit truncation because that width is injective on our small per-instance atom sets, not as a cryptographic bound. On the scored Wikidata 2×2 instances, no HMAC collision was observed. HMAC is computed from the normalized atom string; entity and relation types are not domain-separated, so identical plaintext yields the same token regardless of type. On each scored evaluation instance the normalized entity strings and relation strings are disjoint, so no entity is identified with a relation. The model receives G or $\sigma(G)$ and a query form built from σ (or from English, depending on the arm) and must

return the answer node in that namespace; recovering σ^{-1} is not scored. Sealing is deterministic *given the instance key*; the same plaintext need not map to the same token across prompts.

A 2-hop join factors as two maps we measure separately:

$$\text{selection: } (q, G, \mathcal{C}) \mapsto (r_1, r_2), \quad (1)$$

$$\text{traversal: } (v_0, r_1, r_2, G) \mapsto v_2. \quad (2)$$

In general, traversal is set-valued ($\tau_G: V \times R \times R \rightarrow 2^V$); we evaluate only instances for which $|\tau_G(v_0, r_1, r_2)| = 1$ for the gold plan (and, on two-path items, for the decoy plan as well), so each scored route has a unique tail. Uniqueness is per plan, not “one answer in G .” We write $q \mapsto (r_1, r_2)$ when G and prompt context $\mathcal{C}$ are held fixed. A sealed join plan supplies (r_1, r_2) and tests (2). Unique-path items with v_0 already in q also test (2) when topology leaves only one route. Free sealed English tests (1) then (2) together. Dual-path items hold G fixed, give two complete routes from v_0 , and vary whether demonstrations expose the gold seals. The CEO free-English arm does *not* put v_0 in q as a single graph atom (the question is hashed piecewise), so it is not a pure test of (1).

A 2-hop is *structurally executable* when (r_1, r_2) is given, or when topology leaves only one 2-hop from a start already in q . The prompt *explicitly maps* query semantics onto one relation sequence when it contains English predicates, a legend, or demonstrated seals. When two complete routes exist and relation tokens are arbitrary HMAC strings with no such mapping supplied, selection is *unsupported by explicit cues* under this construction. Residual statistical structure can remain (token form, order, HMAC vs. permutation rates); we do not claim mathematical unidentifiability. Without an external semantic-to-symbol map, the gold route is not explicitly identifiable from the sealed two-path graph alone; any successful selection must rely on residual statistical cues or information supplied elsewhere in the prompt. Low exact-match is therefore unsurprising when the prompt provides no explicit mapping; it is not evidence that traversal failed. Dual-path separates those cases: in-prompt reuse of demonstrated relation-seal correspondences restores end-to-end accuracy; novel seals receive no such mapping. Models may abstain (UNKNOWN) rather than guess. That is selective prediction (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017), studied in language models as calibration and abstention (Kadavath et al., 2022; Wen et al., 2025): coverage is the fraction of items with a non-UNKNOWN parse, and selective accuracy is exact match among those items, not a forced-choice among the two routes.

After sealing, the Eisner unique-path CONTEXT is Michael_Eisner | E334675310f05 | The_Walt_Disney_Company then The_Walt_Disney_Company | Eac04fab538ca | Burbank (12-hex tokens; entities remain English on the primary 2×2).

We use five measurement arms. **Readable English** is the baseline (start in q ; names and predicates unsealed). **Unique opaque** has one 2-hop and no plan: execution when topology uniquely determines the route. **Two-path opaque** has two same-type 2-hops and no plan: end-to-end behavior when two complete routes compete and the prompt supplies no semantic-to-symbol map; the scored output is the answer node or UNKNOWN, not (r_1, r_2) . **Sealed join plan** writes (r_1, r_2) over σ -atoms (execution control); a non-LLM router saturates on the same sequence. **Free sealed English** seals the question and graph with no plan and no legend. On CEO, the question is hashed piecewise, so the full person atom in G typically does not appear in q (confounded; not the matched factorial). Additional scaffolds (name/key legends, sealed demonstrations) appear as ablations.

The scored output on every free arm is that answer node (plaintext city when entities stay English; $\sigma(a)$ when entities are sealed) or UNKNOWN. We infer whether selection was supported from those end-to-end outcomes and from controls (unique topology, written hop, matched vs. novel seals); we do not score an emitted (r_1, r_2) . Throughout, “selection failure” names that inference about the latent (r_1, r_2) stage; we do not observe the pair.

4 Method

Seal recipe. Normalize each atom: strip; collapse whitespace to `_`; map other non-alphanumeric characters (except underscore, period, hyphen) to `_` using the ASCII class `[A-Za-z0-9_.-]` (so Zürich→Z_rich; pure

CJK keeps Unicode); truncate to 64; empty maps to `EMPTY`. The CEO JSON already deleted diacritics upstream. Distinct raw atoms must normalize to distinct strings. Tokens are HMAC-SHA256 over UTF-8 bytes with a per-item 16-byte key; the token is `E` plus the first 6 digest bytes as 12 hex digits. HMAC is keyed renaming, not a security mechanism: 48-bit truncation is instance-wise injective on our small atom sets. On the scored Wikidata 2×2 graphs, normalization and HMAC produced no collisions among prompt atoms. The intervention removes lexical semantics and also changes token-level surface statistics; permutation and alphanumeric generators assess robustness to that latter variation (Table 7). The atom `Michael Eisner` normalizes to `Michael_Eisner`, so those two strings receive the same seal. Hashing the question token-by-token does not reconstruct that graph atom, which is the missing-start failure a name-binding wrapper is designed to prevent. Reported opaque cells are performance under this keyed HMAC construction, not a claim that every opaque identifier distribution yields the same rate.

Matched 2×2 (start in q). The $n=32$ suite is a seeded random sample of 32 of 73 Wikidata-derived CEO records (`random.Random(20260821).sample`; Vrandečić and Krötzsch, 2014), not a published CEO benchmark and not a hand-picked subset. Each instance is 4 triples (unique-path: 6 entities with a disconnected decoy island; two-path: 5 entities sharing the start). The gold two-hop uses `works_at` then the headquarters relation; the question says “led by”. The competing two-path route is synthetic (`ambig_edges` attaches record $i+1$ ’s company/HQ under `partner_of/located_in`; not a Wikidata fact; decoy HQ pool = gold HQ pool). To isolate relation selection from start-entity identification we keep the start string in the question and hold hop length, template, and graph size fixed. Two factors vary: relation labels (English vs. HMAC) and topology (one 2-hop from the start vs. two). Tails are intended as the same type; Wikidata P159 was not filtered (buildings/stations/a street among gold HQs; both OpenAI opaque two-path decoys sit on city/non-city pairs; Appendix A). On Wikidata two-path graphs both intermediates are companies (`works_at/headquartered_in` vs. `partner_of/located_in`). In the primary run the gold 2-hop is listed before the decoy 2-hop in `CONTEXT` (same order in English and opaque). Listing-order robustness is Section 5.2 on the $n=200$ freeze (same OpenAI tag as the named-arm cell). The $n=32$ shuffle is a later snapshot (Appendix A), not that comparison.

The primary two-path prompt is a single user message:

```
MODEL UNDER TEST. Read ONLY this file. Use CONTEXT only. No other files.
Format: ANSWER_PLAIN[...]: <city_or_UNKNOWN>
ARM OPAQUE_AMBIG: opaque relations, English entities;
two 2-hops from start.
```

It does *not* contain the equality-recipe line (Appendix J). The format token is `<city_or_UNKNOWN>`, but six $n=32$ gold HQs are not cities (Appendix A). A model that obeys the format on those items must emit `UNKNOWN`. The powered city-typed cell is $n=200$ (Section 5.1); the $n=32$ lock is the three-family pin. OpenAI English two-path is 31/32 under that `CONTEXT`-only instruction, so the line does not by itself force abstention. Readable names (Eisner, Burbank) still leave OpenAI at `UNKNOWN` 24/32 on opaque two-path: either parametric HQ is treated as inadmissible, or the verb does not map onto one seal. We cannot separate those readings here (Section 7). The CEO three-arm decoy 2-hop begins at a different person; it is not the matched two-path factorial.

Further constructions. The relation-only factorial keeps entities English. A second isolation suite on the same 32 CEO records seals entities, relations, or both, always with the start token in q . Opaque-entity / English-relation (OE) is the ARoG-like cell: readable predicates, anonymous nodes. Both-opaque (WIKI-OO) seals entities *and* relations; it is a complementary ablation, not the primary relation-only 2×2 . Readable entity names therefore leave residual pretrained world knowledge as a possible cue; both-opaque is not a substitute for synthetic identifiers.

WikiMovies facts are sampled from the public MetaQA movie KB (Miller et al., 2016; Zhang et al., 2018). `MOVIE-A32` and `MOVIE-A100` leave English verbs such as *directed* in q ; `MOVIE-B` hashes question atoms except the bracketed start entity. We exclude a genre-vs-language variant because object type would bypass relation selection. This is not an evaluation on the official MetaQA 14K test split. Dual-path holds G fixed with two complete 2-hop routes and varies whether demonstrations expose the gold seals (Appendix D).

Scoring. The format asks for one answer token in the graph’s namespace or UNKNOWN. Primary metric: exact match (UNKNOWN is a miss). Scored UNKNOWN on Wiki-H5/Wiki-OO is an explicit UNKNOWN line. Dual and some no-hop OpenAI cells stored empty completions as UNKNOWN. On two-path arms we also report decoy and UNKNOWN rates (Table 3). Wilson 95% intervals use $z=1.96$ (Wilson, 1927) ($0/32 \Rightarrow [0.00, 0.11]$; $6/32 \Rightarrow [0.09, 0.35]$; $32/32 \Rightarrow [0.89, 1.00]$). Those intervals treat the 32 items as Bernoulli trials on a single draw; they are not a generation-variance interval. OpenAI WIKI-H5 $k=3$ (August lock plus two later draws) is Section 5.1. On the OpenAI Wikidata 2×2 , English two-path is correct and opaque two-path is not on 25 of 32 paired items, and the reverse never occurs (McNemar exact two-sided complete-case $p=5.96 \times 10^{-8}$; McNemar, 1947). That is the two-path paired contrast, not an interaction test, and it largely measures removing the cue that made one sealed route identifiable. The empirical content is the response mode (UNKNOWN vs. decoy) and prompt-surface packing (Section 5.2). That p clears 0.001 (confirmatory McNemar only). **Isolation** means one quiz per file with gold answers stored outside the model-visible prompt (stricter than multi-quiz **batch** files). Isolation $n=32$ uses one quiz per file and a per-item HMAC. Cells marked **n.r.** were not evaluated; we never impute them as zeros. *Mid-hop error* means the model emits the sealed mid-hop company token instead of the headquarters answer.

Models. The primary family is OpenAI **gpt-5.6-sol** (hereafter OpenAI). Composer 2.5 and Grok 4.5 are a robustness note on the same quiz files, not a ranking (Table 3). Secondary packaging probes additionally use an unversioned Claude endpoint (Anthropic model id not stored; Appendix F). **gpt-5.6-sol** is the GPT-5.6 Sol Chat Completions endpoint (Sol is the flagship-equivalent tier of the GPT-5.6 family; <https://developers.openai.com/api/docs/models/gpt-5.6-sol>). OpenAI cells use that API with no tools, temperature omitted because that family rejects temperature 0 (**max_completion_tokens** 512). The runner does not pass **reasoning_effort** and used the unpinned alias **gpt-5.6-sol**; the August lock stores the parsed answer, not the API **model** string. $k=3$ OpenAI draws (August lock plus two 15 September 2026 draws with JSON sidecars that *do* store **returned_model**) are reported in Section 5.1. Composer 2.5 and Grok 4.5 ran through Cursor’s agent SDK on the same quiz files, with a reply-only preamble OpenAI never received. Tools, retry, and empty-reply handling are not equated (OpenAI locked empty completions as UNKNOWN). Unique-path opaque stays high on all three (Composer 27/32, OpenAI 32/32, Grok 30/32): those SDK cells still execute unique-path joins. Named-arm difference-in-differences (DiD) is most readable for OpenAI (Grok English two-path 4/32 vs. English, no ARM, hashed ids 24/32; Section 5.2). Unique both-opaque 32/32 is the three-family traversal cell. Isolation quizzes (**runs/**) and locked exact-match scores (**results/**) are in the supplementary archive. Inference settings are Appendix A.

5 Results

Table 2 names the cells. WIKI-H5 is the Wikidata CEO relation-only 2×2 with the start already in q (OpenAI primary; Composer/Grok are a different harness). WIKI-OO seals entities *and* relations on the same graphs; OE seals entities only. MOVIE-A32 is WikiMovies $n=32$ with English verbs still in q ; MOVIE-A100 is the $n=100$ version; MOVIE-B hashes those verbs except the start. Header $\times$ decoy is the same 32 people with hashed case ids, crossing the ARM line with cyclic vs. constant decoys (Section 5.2). MOVIE-B is a cross-dataset check, not a resampling of WIKI-H5. Packaging probes ($n \leq 16$) are Appendix G and do not bear on the 2×2 .

5.1 Opacity $\times$ path ambiguity

WIKI-H5 holds the start string in q and varies only relation lexicality (English vs. HMAC) and the number of 2-hops from that start (one vs. two). OpenAI is the primary family. Composer 2.5 and Grok 4.5 are Table 3 robustness, not a ranking; they used Cursor’s agent SDK with a preamble OpenAI never received.

City-typed $n=200$ (OpenAI). The $n=32$ source file has 73 CEO rows and six non-city P159 golds. WIKI-H5-200 is a new freeze: 200 unique people and companies, HQ frequency capped at 3, every HQ an instance of a city-like Wikidata class, person/company/HQ QIDs stored, and all 200 P169+P159+P31 triples re-checked live against the Query Service (**wikidata_ceo_hops_n200.json**). It does not overwrite the $n=32$ lock. OpenAI named-arm: unique English and unique opaque 200/200, English two-path 199/200, opaque

Table 2: Cell inventory, listed before results. WIKI-H5 is the OpenAI discovery (Composer/Grok in Table 3, not a ranking). WIKI-H5-200 is the powered city-typed QID freeze (OpenAI). Header×decoy is Section 5.2. Six $n=32$ P159 golds are not cities (Appendix A). A32/A100/B share the actor→movie→director vs. writer template; A100 Condition A uses a thinner ARM than A32. Isolation cells use per-item HMAC; batch three-arm $n=12/n=24$ share one key per file.

ID	Dataset	Condition	n	Models	Scored
WIKI-H5	Wikidata CEO	rel-only 2×2, start in q	32	OpenAI (primary)	node
WIKI-H5-200	Wikidata CEO, city P31	same 2×2, QID freeze	200	OpenAI	node
Header×decoy	same 32 people	ARM line × decoy, hashed ids	32	3 families	node
Header×decoy-200	same 200 people	ARM × decoy, hashed ids	200	OpenAI	node
WIKI-OO	same graphs	both-opaque	32	3 families	$\sigma(a)$
WIKI-OO-200	$n=200$ freeze	both-opaque	200	OpenAI	$\sigma(a)$
WIKI-SHUF	same two-path	listing order	32	OpenAI	city
MOVIE-A32	WikiMovies	actor→movie→dir vs writer	32	3 models	person
MOVIE-A100	WikiMovies	same hops; verbs in q	100	OpenAI	person
MOVIE-A200	WikiMovies	$n=100$ prefix + 100 new actors	200	OpenAI	person
MOVIE-B	WikiMovies	same hops; verbs hashed	100	OpenAI	person
MOVIE-B200	WikiMovies	Condition B on $n=200$ freeze	200	OpenAI	person
DUAL	WikiMovies/synth	matched vs. novel demos	32	3 models	$\sigma(a)$
DUAL-200	WikiMovies $n=200$	matched vs. novel	200	OpenAI	$\sigma(a)$
CEO-NL	Wikidata CEO	piecewise q , no plan	32	3 models	$\sigma(a)$
CEO-NL-200	$n=200$ freeze	piecewise q , no plan	200	OpenAI	$\sigma(a)$

two-path 38/200 (UNKNOWN 151, decoy 11; Wilson [0.14, 0.25]), plan 200/200. The 38/200 rate sits inside the $n=32$ Wilson band for 6/32. Decoy 0 on the $n=26$ city slice was a small-sample artifact: $n=26$ cannot resolve a $\sim 5\%$ decoy rate. At $n=200$ the usual reply is still UNKNOWN ($\approx 14:1$ vs. decoy). Prompt-surface packing of this freeze is Section 5.2. $n=200$ is OpenAI-only and one draw; three-family cells stay at $n=32$.

$n=32$ **discovery**. On unique sealed routes OpenAI executes (opaque unique 32/32; plan 32/32). On competing sealed routes the usual named-arm reply is UNKNOWN, not the decoy. On 26 city-valued items unique English, unique opaque, and English two-path are 26/26; named-arm opaque two-path is 5/26, 2/26, 2/26 across three draws. August: UNKNOWN 21, decoy 0, selective accuracy 5/5. The lock is opaque two-path 6/32 (UNKNOWN 24, decoy 2); both decoys sit on non-city P159 golds. Wilson: 6/32 $\Rightarrow$ [0.09, 0.35]; city slice 5/26 $\Rightarrow$ [0.09, 0.38]. Figure 1 plots the $n=32$ 2×2.

Let Δ_{opq} be opaque minus English exact-match on $n=32$: unique +1/32, two-path −25/32 (25 English-correct and opaque-wrong, 0 reverse; McNemar $p=5.96 \times 10^{-8}$). Opacity is not a randomized within-item treatment (Appendix I).

OpenAI $k=3$. Two 15 September draws of the unpinned alias (gpt56k3r2/r3) are a new snapshot, not a generation-variance interval. Named-arm opaque two-path is 6/32, 3/32, 3/32; Composer 10, 7, 6 and Grok 1, 0, 0 move the same way. Same-day draws 2 and 3 match; August is highest in all three families. The $n=32$ same-date case-id pair is 11 September named-arm 5/32 vs. hashed-id H5-cyclic 20/32 (Section 5.2). Named-arm English two-path is OpenAI 31/32, Composer 26/32, Grok 4/32: a packaging×family effect, not a ranking. Opaque unique stays high (Composer 27/32, Grok 30/32); a written hop recovers (28/32, 32/32). Two-path coverage / selective accuracy (named-arm): OpenAI opaque 8/32, 6/8; Composer 10/32, 10/10; Grok 1/32, 1/1. Two other generators still have unique > two-path (Table 7).

5.2 Prompt surface as a factor

The named-arm WIKI-H5 prompt contains this line:

ARM OPAQUE_AMBIG: opaque relations, English entities; two 2-hops from start.

Table 3: Four-way outcomes: gold / decoy / UNKNOWN / other wrong token. OpenAI is the primary family (August lock; $k=3$ in the surrounding text). Composer 2.5 and Grok 4.5 used Cursor’s agent SDK with a reply-only preamble OpenAI never received; this table is not a ranking. On this table UNKNOWN is an explicit UNKNOWN line (missing/unparsed 0; both-named 0). Composer “other” is truncated gold Lod (English unique, opaque unique, and plan) plus one wrong city (Fox_Plaza→Los Angeles). OpenAI’s two decoys are P159 type-asymmetric pairs. Coverage and selective accuracy for every two-path row are in the surrounding text. Grok named-arm English two-path is mostly UNKNOWN.

Wikidata named-arm (iso. $n=32$)	OpenAI	Composer 2.5	Grok 4.5
English, unique	31/0/1/0	29/0/2/1	32/0/0/0
English, two paths	31/0/1/0	26/0/6/0	4/0/28/0
Opaque, unique	32/0/0/0	27/0/3/2	30/0/2/0
Opaque, two paths	6/2/24/0	10/0/22/0	1/0/31/0
Opaque two paths + plan	32/0/0/0	28/0/3/1	32/0/0/0

That sentence tells the model in English that two 2-hops exist. It is an uncontrolled treatment if abstention is attributed only to opacity-meets-ambiguity. Prompt-surface packing is a factor (Sclar et al., 2024). Two manipulations must not be collapsed.

Case-id wording, ARM line held fixed. The case id appears twice: in Format: ANSWER_PLAIN[id] and in the ID line. Named-arm uses F22_OPAQUE_AMBIG_i; OPQ_H5_CYC substitutes hashes HDA_*. ARM line and CONTEXT are identical. On the city-typed $n=200$ freeze, the same 512-token contrast (named-arm vs. hashed-id H5-cyclic; tags gpt56n200 and gpt56n200h512) is 38/200 vs. 134/200 (+96 items; H5-cyclic has one NO_OUTPUT). The $n=32$ discovery pair is same-date 11 September: named-arm gpt56sep 5/32 vs. hashed-id H5-cyclic gpt56ab1512 20/32 (+15; one NO_OUTPUT each). Both prompts keep the ARM line, so this is not a test of whether the prompt announces ambiguity. $k=3$ hashed-id gold is 20, 21, 22; against named-arm 6/3/3 only +18 and +19 are same-day.

ARM line, case ids held hashed. Header×decoy at 4096/medium on the $n=200$ freeze is Table 4: cyclic/constant no ARM 174/151, H5 129/44, K2 179/144; English no ARM cyclic 197/200. H5 cyclic vs. constant McNemar 95 vs. 10, $p=1.6\times 10^{-18}$; no-ARM 30 vs. 7, $p=1.9\times 10^{-4}$. The $n=32$ discovery (Table 8) already showed the H5 interaction (cyclic 20 vs. constant 10; McNemar 14 vs. 4, $p=0.031$) and not no-ARM (gold 25 vs. 27; McNemar 1 vs. 3, $p=0.625$). K2 is a shorter ARM that still names two hops. K2 and no-ARM were not run at 512.

Listing order. A shuffled CONTEXT listing of the same $n=200$ freeze and HMAC keys, same OpenAI tag as the primary named-arm cell, scores opaque two-path 59/200 against 38/200 for the gold-first listing (+21 from reordering). Gold-first is 25/100, decoy-first 34/100; a two-proportion test is $p\approx 0.16$ (not significant). The $n=32$ shuffle is a later snapshot (Appendix A) and was too small to read a decoy-first effect (decoy 0/16 vs. 1/16).

Table 4: Header×decoy on the city-typed $n=200$ freeze (same people as WIKI-H5-200). English, no ARM, cyclic is 197/200. H5 cyclic vs. constant McNemar 95 vs. 10 ($p=1.6\times 10^{-18}$). Same-date 512-token hashed-id H5-cyclic is 134/200 (one NO_OUTPUT). $n=32$ golds: Table 8.

ARM line (OpenAI, hashed ids, 4096/medium, $n=200$)	Cyclic	Constant
no ARM	174/200	151/200
H5	129/200	44/200
K2	179/200	144/200

English Grok. Named-arm English two-path is Grok 4/32 (UNKNOWN 28). The English header×decoy cell (ENG_NONE_CYC) is 24/32. That cell has no ARM line and hashed ids, so two things differ from the 4/32

Table 5: WIKI-OO and OE on the same 32 graphs (start in q). OpenAI OE two-path 32/32; both-opaque two-path 1/32. Plan is a control.

Arm	Ent.	Rel.	OpenAI	Composer	Grok
Unique path	opaque	Eng.	32/32	32/32	32/32
Two paths	opaque	Eng.	32/32	28/32	3/32
Unique path	opaque	opaque	32/32	32/32	32/32
Two paths	opaque	opaque	1/32	0/32	0/32
Two paths + plan	opaque	opaque	32/32	28/32	32/32

named-arm English prompt. Call it English, no ARM, hashed ids—not “arm-neutral English.” A prompt-verbatim Grok rerun of the named-arm English files (**grok45pn**) stays 4/32, so the Cursor preamble is not that gap. OpenAI named-arm English is 31/32 vs. English, no ARM, hashed ids 29/32: OpenAI English does not abstain either way. MOVIE-B topology-only two-path is 3/100; an archived banner that names the two routes is 25/100 (a different surface, not the Wiki-H5 case-id contrast).

What the data support: exact-match under competing sealed routes is strongly sensitive to prompt surface form. At $n=200$, same-budget 512-token case-id wording moves cyclic gold by 96 items (38/200 vs. 134/200). At 4096/medium, ARM on *constant* decoys moves gold by 107 items (no ARM 151 vs. H5 44); on cyclic, K2 vs. H5 is 50 items (179 vs. 129). Budgets differ, so these are not a ranking of manipulations; the ARM effect is not a small add-on to case-id.

A consequence for the contrast with Guo et al. (2025). That work reports collapse to guessing among end entities when distractor two-hops have other sources. Our named-arm OpenAI cell is abstention (UNKNOWN 24/32, decoy 2/32 at $n=32$; UNKNOWN 151, decoy 11 at $n=200$), not a coin-flip onto the decoy. That “UNKNOWN rather than decoy” pattern is not a three-family law. Grok no-ARM opaque commits are 6 gold / 3 decoy cyclic and 1 / 5 constant (7 vs. 8 pooled): chance. For Grok, preferring UNKNOWN over the decoy is a named-arm phenomenon.

Composer/Grok hashed-id H5-cyclic cells used **-preamble none** and score 6/32 and 0/32; the named-arm locks used the Cursor preamble and score 10/32 and 1/32. Both families go *down* with hashed ids, a sign reversal relative to OpenAI. We did not re-query Composer named-arm with the prompt file verbatim, so that reversal mixes case-id wording with preamble deletion and is not interpretable. Grok is at floor on both opaque two-path cells.

5.3 Both-opaque and readable relations

Table 5 is a second isolation suite on the same 32 records (**ANSWER_SEALED**; start seal in q). Unique-path both-opaque is 32/32 for all three families. Composer/Grok relation-only unique (27/32, 30/32 in Table 3) is a different file, not evidence that hiding entities helps. Two-path both-opaque is OpenAI 1/32 (Wilson [0.01, 0.16]), Composer 0/32, Grok 0/32. A written hop recovers (32/32, 28/32, 32/32). OpenAI OE two-path (opaque entities, English relations) is 32/32: readable predicates restore end-to-end accuracy with anonymous nodes under this construction. Grok OE two-path remains 3/32. On the $n=200$ freeze (OpenAI), unique both-opaque and OE two-path are 200/200; both-opaque two-path is 12/200.

WikiMovies unique-path with English verbs in q (MOVIE-A32) is 32/32 for all three families; opaque two-path is 17/32, 17/32, 0/32, with plan 32/32. MOVIE-A32 and MOVIE-A100 are the same actor→movie→director vs. writer template (person tails). A32’s ARM names two 2-hops; A100’s is **OPAQUE_AMBIG** $n=100$. On 32 A32 items matched by gold name (31 unique names) OpenAI is 17/32 vs. 29/32 under A100’s thinner ARM. The extra 68 items are 67/68, which is why the $n=100$ cell is 96/100. Keys differ; they cannot explain an English-name lookup. Director is always the first person-tail, so parametric knowledge and a listing-slot heuristic are not separated. Movie-B keeps those names, hashes the verbs, and is 3/100. The contrast is not a same-prompt Wiki-H5 replication; it remains a header-width comparison on WikiMovies (Section 5.2). Verbs in q can cue seals; we do not treat Condition A as a selection result.

Table 6: MOVIE-A100/MOVIE-B: same actor→movie→director vs. writer template as MOVIE-A32 (Miller et al., 2016; Zhang et al., 2018) (OpenAI $n=100$, start in q). Condition B: unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; coverage 13/100, selective accuracy 3/13).

WikiMovies people (OpenAI, $n=100$)	Unique	Two-path
English relations, Condition A (verb in q)	100/100	100/100
Opaque relations, Condition A (verb in q)	100/100	96/100
Opaque relations, Condition B (verbs hashed)	100/100	3/100
Opaque two-path + plan, Condition B	—	100/100

5.4 WikiMovies: hashed verbs

Table 6 uses the same actor→movie→director vs. writer template as MOVIE-A32 (person-valued tails). Condition A leaves *directed/starred* in q (not a selection result). Condition B hashes those verbs: OpenAI unique 100/100, two-path 3/100 (UNKNOWN 87, decoy 0, both-named 10; Wilson [0.01, 0.08]). Coverage is 13/100 and selective accuracy 3/13. A written hop is 100/100. Composer/Grok Condition A two-path: 62/100, 36/100; Condition B $n=100$ is OpenAI-only. On the $n=200$ freeze, Condition A two-path is 122/200 (Wiki-H5 topology ARM, not A100’s ARM OPAQUE_AMBIG $n=100$ banner, so not a resample of 96/100); Condition B is 7/200. Movie-B’s ARM names two 2-hops (Wiki-H5-style); A100 Condition A does not. The $n=32$ MetaQA-style template cell (Appendix H) is not MOVIE-B.

5.5 In-prompt seal reuse

Sealed demonstrations enable the corresponding joins when the relevant relation seals appear in the demos. Dual-path probes hold G fixed with two complete 2-hop routes (gold vs. decoy) and vary whether those gold seals appeared in the demonstrations. Matched-relation demos expose the gold relation seals; novel-relation demos teach seals that do not appear on the gold route. An explicit join plan is the execution upper bound on the same graphs. “Asymmetric” noise makes the decoy look cleaner; “balanced” noise is symmetric on both mids. Matched-relation is in-prompt reuse of demonstrated correspondences; it is not induction of a general cipher.

Isolation $n=32$ (Figure 2, Appendix D): OpenAI saturates matched-relation under both noise regimes (32/32, decoy 0). Novel-relation balanced is exact match 0/32: all 32 replies are UNKNOWN (24 empty completions). Novel-relation asymmetric is also 0/32 (UNKNOWN 24, of which 17 empty, decoy 8). An explicit dual-path plan scores 32/32. WikiMovies director/writer Dual is 32/32 vs. 0/32 for all three families (dualpath_wikimovies_iso_*; Composer novel decoy 2/32), with plan 32/32. On the synthetic Dual suite, Composer isolation-tag novel-relation UNKNOWN (composer25) is a sealed-demo helper; free-form composer25ff is decoy 32/32. Grok free-form (grok45ff) novel-relation asymmetric is UNKNOWN on all 32 items. Last-demo answer-token copy is 0/32 on every arm. On the $n=200$ freeze (OpenAI), matched is 200/200 and novel is 2/200, with plan 200/200.

On a two-path graph a forced uniform guess among the two complete routes is 1/2; that null does not apply once UNKNOWN is a third action. The pattern is consistent with prompt-local reuse of demonstrated seals and provides no evidence of systematic transfer to unseen relation tokens. We do not interpret this as a variable-binding mechanism. Arm definitions: Appendix D.

5.6 Missing-start control

The CEO isolation three-arm is an execution upper bound plus a missing-start cell, not the isolation of selection. Join plans score 32/32 on both plaintext and sealed atoms; SealRouter given the gold sealed relation sequence is 32/32; free sealed English scores 0/32 (all UNKNOWN) for Composer 2.5, OpenAI, and Grok 4.5 (Table 11). On the $n=200$ freeze, plan remains 200/200 and free sealed English 0/200. We exclude a batched free-sealed cell that can succeed by returning a unique neighbor without using q .

6 Discussion

A join is structurally executable when the route is unique or (r_1, r_2) is given (OpenAI unique opaque 32/32 and 200/200; both-opaque unique 32/32 on three families; written hop OpenAI/Grok 32/32, Composer 28/32). The prompt explicitly maps query semantics onto one relation sequence when English predicates, a legend, or demonstrated seals are present. When two complete routes exist and that mapping is absent, the usual named-arm reply is UNKNOWN rather than the decoy (151 vs. 11 at $n=200$, about 14:1; Table 3 is the three-family discovery). We do not claim mathematical unidentifiability. Exact-match also moves with prompt surface (Section 5.2): at $n=200$, 512-token case-id wording is 38/200 vs. 134/200 on cyclic decoys; at 4096/medium the ARM line is non-monotonic and interacts with decoy type (Table 4; H5 McNemar 95 vs. 10). A shuffled listing of the same freeze adds 21 golds (59/200 vs. 38/200). English relations restore OpenAI with anonymous entities (32/32); Grok still abstains (3/32). Dual-path reuses demonstrated seals; novel seals do not transfer (2/200 novel vs. 200/200 matched). Unique topology with the start in q , English predicates, matched demonstrations, or a same- σ equality recipe restore execution; a bare schema list does not (Table 13). A high English two-path rate does not imply the same join once relations are opaque. OIR diagnoses residual join ability after readable relations and retrieved paths are withheld; PrivGemo keeps those paths and answers locally (Tan et al., 2026).

7 Limitations

OpenAI is the primary family. The $n=200$ freeze is OpenAI-only and one draw; Composer/Grok stay at $n=32$ (Table 3 is not a ranking). Decoy 0 on the $n=26$ city slice was underpowered for a $\sim 5\%$ decoy rate; $n=200$ is decoy 11 vs. UNKNOWN 151. The $n=32$ format token makes six P159 golds ill-posed; $n=200$ does not use those golds. The $n=32$ suite is 32 of 73 records, one template, one seed; decoys are cyclic, so Wilson intervals are slightly optimistic (McNemar is not). $k=3$ mixes the August lock with later alias draws (Section 5.1): same-day draws match, August is high in all three families, and this is not a generation-variance CI. English entities leave parametric cues; HMAC two-path rates vary by generator (6/32, 15/32, 12/32). Selection is inferred from the answer node. Named-arm selective accuracy stays high; Grok no-ARM opaque commits are near chance (Section 5.2). Header $\times$ decoy at $n=200$ is Table 4; Table 8 is the $n=32$ discovery. K2 and no-ARM were not run at 512. MOVIE-A32 vs. MOVIE-A100 (17/32 vs. 29/32) is a thinner ARM on WikiMovies; $n=200$ Condition A uses the Wiki-H5 topology ARM, not that banner. Dual novel OpenAI UNKNOWN includes empty completions. Forced-choice, scored (r_1, r_2) , and an HQ-city swap are not in this evaluation. The ARM line was not randomized independently of case-id wording on the original lock. Composer hashed-id H5-cyclic used **-preamble none**; the named-arm Composer lock used the Cursor preamble, so that sign reversal is uninterpretable.

Broader Impact Statement

Public Wikidata CEO records (CC0; Vrandečić and Krötzsch, 2014) and the WikiMovies/MetaQA graph as released (Miller et al., 2016; Zhang et al., 2018). Appendix cells use 2WikiMultihopQA (Apache-2.0; Ho et al., 2020) and Spider (CC-BY-SA-4.0; Yu et al., 2018). No personally sensitive information. Replies are hash-pinned, not a training dump. OIR is a diagnostic, not a confidentiality mechanism (English names remain on the primary 2×2).

8 Conclusion

Unique-route sealed execution succeeds with the start in q (opaque unique OpenAI 32/32 and 200/200; both-opaque unique 32/32 on three families). On the city-typed $n=200$ freeze, named-arm opaque two-path is 38/200 (UNKNOWN 151, decoy 11); hashed-id H5-cyclic at 512 tokens is 134/200 (one NO_OUTPUT). The $n=26$ decoy-0 slice was too small to see that rare decoy rate. Exact-match under two sealed routes is strongly sensitive to prompt surface: case-id wording moves 96 cyclic golds at 512 tokens; at 4096/medium, ARM on constant decoys moves 107 golds (Table 4). Grok no-ARM opaque commits are near chance, so “UNKNOWN rather than decoy” is not a three-family law.

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A Protocol, scoring, and models

Isolation quizzes (`runs/`) and locked scores (`results/`) are in the supplementary archive (Python ≥ 3.10). OpenAI cells: `gpt-5.6-sol`, no tools, August 2026 lock plus two 15 September 2026 $k=3$ draws on the same alias, temperature omitted, `max_completion_tokens` 512; `reasoning_effort` was omitted. The August runner stores the parsed answer, not the API `model` string or `usage` (including reasoning-token counts); $k=3$ sidcars store `returned_model`. WIKI-SHUF is a 7 September 2026 snapshot on the same 32 keys. The $n=200$ listing shuffle uses the same OpenAI tag as named-arm `gpt56n200` (not a new snapshot). Composer 2.5 and Grok 4.5 used the same quiz files through Cursor’s agent SDK with a reply-only preamble OpenAI never received; settings are not equated. Composer Wiki-OO unique/two-path, Composer/Grok CEO, Dual `*ff` and some Dual isolation-tag arms are answer-line locks (cite `*ff`; Grok Wiki-OO is a one-line tail). A later Composer Wiki-OO unique transcript rerun (`composer25tx`) is 32/32; it does not replace the August lock. Two-path stays 0/32 (one decoy vs. lock UNKNOWN 32). Some Dual and no-hop OpenAI cells store empty completions as UNKNOWN; Wiki-H5/Wiki-OO do not. WIKI-H5 uses seed 20260821 and a per-item 16-byte key derived

from `oir-fact-2x2-iso-v1` and the item index. The 32 records are `random.Random(20260821).sample` from the 73-row CEO file; two-path decoys attach record $i+1 \bmod 32$, so each item’s decoy company/HQ is the next item’s gold. Wilson intervals in the body treat items as i.i.d.; that is slightly optimistic under this cycle. WIKI-H5-200 is a separate freeze (`data/real/wikidata_ceo_hops_n200.json`, seed 20260915): company P169 person, company P159 HQ, HQ P31 a city-like class (not station/building/state), unique people and companies, at most three rows per HQ city, cyclic decoys arranged so consecutive items do not share HQ or person. Each row stores Wikidata QIDs. Live SPARQL re-check of all 200 P169+P159 pairs and city P31 is `results/wikidata_ceo_hops_n200_verify.json` (200/200). HMAC keys are `oir-fact-2x2-iso-n200-v1`; $n=32$ prompts are not rewritten. Header×decoy, Wiki-OO, shuffle, and CEO-NL $n=200$ use that freeze. WikiMovies $n=200$ keeps the $n=100$ people prefix and adds 100 new unique actors from `wiki_entities_kb.txt`. OpenAI $n=200$ tag is `gpt56n200` (header 4096/medium: `gpt56n200ab1`; 512-token H5-cyclic: `gpt56n200h512`, gold 134, UNKNOWN 44, decoy 20, other 1, `NO_OUTPUT 1`). A SHA-256 lockfile over git-tracked JSON summaries and per-item reply files in `results/` is `results/SHA256SUMS` (regenerate with `harness/write_sha256sums.py`; ignored packs such as `longctx_1m*` are omitted). That file is a hash pin, not a timestamped preregistration. Verify with `shasum -c results/SHA256SUMS`. Cite the supplementary lockfile `results/SHA256SUMS` rather than a mutable checkout. No models were trained. OpenAI cells used Chat Completions on CPU-side API calls (`max_completion_tokens 512`; header×decoy 4096). WIKI-H5 isolation is 32 items × five named arms; header×decoy is 224 queries per family. Composer 2.5 and Grok 4.5 used Cursor’s hosted agent API on the same quiz files. Wall-clock is API latency, not local GPU-hours. WIKI-OO is a second suite on those records (`oir-entrel-2x2-iso-v1` and index). MOVIE-A32/MOVIE-A100/MOVIE-B and Dual isolation items are per-item keys; Dual demonstrations reuse that item’s key. CEO-NL isolation $n=32$ uses seed 20260820 and per-item keys. The $n=12$ three-arm is **batch** with one shared key (`oir-ceiling-three-arm-v1`, seed 20260728) for Composer 2.5, OpenAI, and Grok 4.5; $n=24$ is the same protocol at a larger n . We never impute n.r. as zeros.

P159 gold HQs that are not cities: `Broadway`, `NRK_Marienlyst`, `Fox_Plaza`, `CNN_Center`, `Lod_Railway_Station`; `New_Jersey` is a state. OpenAI opaque two-path decoys are items 27 and 31 (`CNN_Center` vs. `Swindon`; `Lod_Railway_Station` vs. `Burbank`). Both decoys are on the non-city subset; city-valued OpenAI opaque two-path is gold 5/26, decoy 0, UNKNOWN 21 (too small for a ~5% decoy rate; $n=200$ is decoy 11, UNKNOWN 151). Every non-two-path miss in Table 3 for all three families falls on those six golds. Composer “other” on that table is truncated `Lod` (item 31) plus `Fox_Plaza`→Los Angeles. Source slice: `Kansas_City` 10/73, `Burbank` 6/73; Disney three times in the 32. An archived Movie-B ARM banner naming director vs. writer is OpenAI two-path 25/100; the locked topology-only Condition B header is 3/100.

The CEO three-arm graphs add extra edges on the gold company (owned-by, partner-of, meta-of). One question template is used. Free English hashes the question piecewise, so the full person atom typically does not appear in q .

B Generator robustness

Table 7: Generator robustness on the same 32 Wikidata graphs and question template. Unique-path accuracy exceeds two-path accuracy under three independently generated opaque relation maps; the two-path *rate* is not invariant. Wilson: 29/32 $\Rightarrow$ [0.76, 0.97], 12/32 $\Rightarrow$ [0.23, 0.55], 15/32 $\Rightarrow$ [0.31, 0.64]. Alphanumeric two-path is all abstention or gold (decoy 0).

Opaque generator (OpenAI, start in q)	Unique	Two-path
Keyed HMAC (E+12hex)	32/32	6/32
Random permutation (E+12hex)	31/32	15/32
Random alphanumeric (not HMAC-shaped)	29/32	12/32

A random permutation of relation labels (HMAC-shaped tokens, not keyed HMAC) on the same Wikidata 2×2 preserves unique-path accuracy exceeding two-path accuracy: OpenAI 31/32 vs. 15/32 (plan 30/32); Composer 28/32 vs. 2/32 (plan 28/32); Grok 31/32 vs. 0/32 (plan 32/32).

An English-entity ambiguity sweep $k=1..5$ still seals relations (HMAC key `oir-ambig-curve-iso-v1`) and, for every item, reuses one constant decoy route (Cogent Communications $\rightarrow$ Washington, DC). OpenAI stays high (29/32, 29/32, 28/32, 27/32, 28/32; plan 32/32); Composer declines (27/32 to 11/32; plan 29/32); Grok is 29/32 at $k=1$ then 0/32 for $k \geq 2$ (plan 32/32). K2 on the ambig-curve suite (curve HMAC key, constant decoy, named K2 header) is OpenAI 29/32 vs. Wiki-H5 6/32; that is not the header $\times$ decoy K2 cells (30/32 cyclic, 26/32 constant). Names such as Disney remain, so k is not isolated from entity semantics; neither curve is a replication of both-opaque 1/32.

Header $\times$ decoy uses arm-neutral case ids on the same 32 people (`header_decoy_ablation_iso_*.json`; body Section 5.2). Cells are named by header and decoy: H5 keeps the Wiki-H5 ARM line, K2 keeps a shorter two-hop ARM, and “no ARM” deletes the line; cyclic decoys attach record $i+1$, constant decoys reuse Cogent Communications $\rightarrow$ Washington, DC. OpenAI 4096/medium $n=32$ golds:

Table 8: Header $\times$ decoy gold counts on the same 32 people. English, no ARM, cyclic is 29/32. H5 McNemar 14 vs. 4, $p=0.031$; no-ARM gold 25 vs. 27, McNemar 1 vs. 3, $p=0.625$. Body Table 4 is the $n=200$ lock.

ARM line (OpenAI, hashed ids, 4096/medium)	Cyclic	Constant
no ARM	25/32	27/32
H5	20/32	10/32
K2	30/32	26/32

Composer and Grok the same day, Cursor SDK, prompt file verbatim (`-preamble none`), 224/224 finished: H5 6/2 and 0/0; K2 20/13 and 0/0; no ARM 21/18 and 6/1; English 26 and 24 (markdown wrappers on four Composer answers stripped at score time). A named-arm English Grok rerun with the prompt file verbatim (no Cursor preamble; tag `grok45pn`) stays 4/32 (rate, not item identity): the Cursor preamble is not that 4/32 cell. The 24/32 English cell (`ENG_NONE_CYC`) has no ARM line *and* hashed ids, so two things differ from named-arm English 4/32; do not read it as a pure case-id contrast. English, no ARM, hashed ids vs. no-ARM cyclic opaque is Grok 24/32 vs. 6/32 (McNemar 18 vs. 0, $p=7.63 \times 10^{-6}$); OpenAI 29 vs. 25 and Composer 26 vs. 21 are underpowered at $n=32$. Grok no-ARM opaque commits are 6 gold / 3 decoy cyclic and 1 / 5 constant (7 vs. 8 pooled). Named-arm OpenAI snapshot 5/32 (512 tokens, one `NO_OUTPUT`) vs. H5 cyclic at the same budget 20/32 (also one `NO_OUTPUT`) is the same-date case-id pair in Section 5.2. Constant decoy does not raise the Wiki-H5-header cell on any family.

C Hypothesis status and maps

Table 9: WIKI-H5 maps on the $n=32$ city-valued discovery slice (OpenAI unless noted). Exact match on the answer node, not (r_1, r_2) . August opaque two-path 5/26; lockfile 6/32; powered freeze 38/200 (Section 5.1). Plan is 26/26 on this slice (32/32 on $n=32$). Last row is CEO-NL ($n=32$).

Arm	Input to the model	Maps tested	Score
English, unique	readable q , start in q	e2e: (1)+(2)	26/26
English, two paths	readable q , start in q	e2e: (1)+(2)	26/26
Opaque, unique	start in q ; one route; no plan	traversal (2) (e2e)	26/26
Opaque, two paths	start in q ; two routes; no map	e2e (selection unobserved)	5/26
Plan (control)	opaque q plus (r_1, r_2)	traversal (2)	26/26
Binding fail ($n=32$)	piecewise q ; start usually absent	start-ID (confounded)	0/32

Table 10: Hypothesis status. H4 exact-match is OpenAI Dual; Composer free-form novel is decoy 32/32 (composer25ff). H5 is named-arm WIKI-H5 (OpenAI). H6 is MOVIE-B. H7 is OpenAI-specific (OE vs. WIKI-OO). H8 is prompt surface on the same graphs (Section 5.2).

Hypothesis	Test	Status
H1. Opacity alone blocks traversal	unique-path, start in q	not supported (32/32)
H2. Written hop restores e2e (control)	two-path + plan; SealRouter	control (32/32)
H3. Input-binding control (CEO piecewise q)	CEO isolation NL	control (0/32); not selection
H4. Matched demos restore shown seals	dual-path matched vs. novel	OpenAI 32/32 vs. 0/32
H5. End-to-end accuracy degrades when opaque relations meet path ambiguity	matched 2×2 , start in q	named-arm 25 vs. 0 (city 21 vs. 0); Sec. 5.1
H6. Movie KB, hashed verbs (check)	WikiMovies Cond. B $n=100$	OpenAI two-path 3/100
H7. Readable relations with opaque entities	Wikidata OE vs. OO	supported for OpenAI here
H8. Prompt surface moves two-path exact match	case-id at 512 cyclic; ARM $\times$ decoy at 4096	$n=200$ case-id +96; H5 McNemar 95 vs. 10; Sec. 5.2

Table 11: Missing-start control, not WIKI-H5. Batch $n=12/n=24$ and packaging rows: Appendix G.

CEO-NL isolation $n=32$	Composer 2.5	OpenAI	Grok 4.5
Sealed plan / SealRouter	32/32	32/32	32/32
Free sealed English (piecewise q)	0/32	0/32	0/32

Table 12: Three-family isolation beyond Table 3. MOVIE-A32 leaves verbs in q (not a selection result). Both-opaque is Table 5 compactly.

Cell (iso. $n=32$)	OpenAI	Composer 2.5	Grok 4.5
MOVIE-A32 opaque unique	32/32	32/32	32/32
MOVIE-A32 opaque two-path	17/32	17/32	0/32
MOVIE-A32 two-path + plan	32/32	32/32	32/32
Both-opaque unique	32/32	32/32	32/32
Both-opaque two-path	1/32	0/32	0/32
Both-opaque two-path + plan	32/32	28/32	32/32

D Dual-path arms

We name the arms by what the demos teach. **Matched-relation** (asymmetric or balanced noise): demos expose the gold relation seals; the decoy path uses different seals. **Novel-relation** (asymmetric or balanced noise): demos teach seals that do not appear on the gold route; both routes are novel at quiz time. **Explicit join plan**: gold relation sequence is written in the prompt.

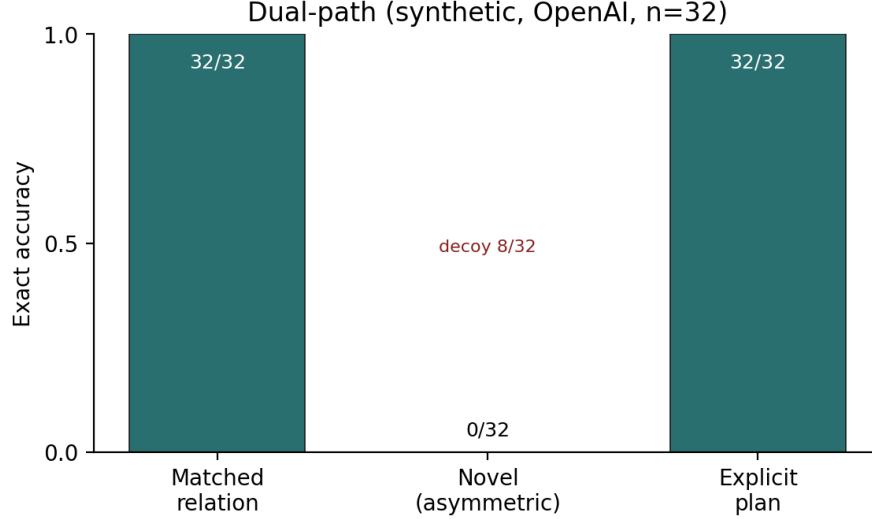


Figure 2: DUAL (synthetic, OpenAI $n=32$): matched 32/32; novel 0/32 (asymmetric decoy 8); explicit plan 32/32.

An $n=8$ cell is zero demos 2/8; one or more demos 8/8. Isolation $n=32$ is the reported dual-path measurement. Local `mistral/qwen35` runs are not in the roster (matched-asymmetric 0/32; Qwen matched-balanced 7/32; plan n.r.). Composer isolation-tag novel-relation UNKNOWN is `adv_induction_n32_iso_composer25.json` (helper `run_induce_mut_n32_iso.py`); the free-form lock is `composer25ff` (decoy 32/32). Grok free-form is `grok45ff`.

E No-hop binding (JSON)

Table 13: Selection support without a written hop (Wikidata CEO JSON; city or mentor country is the distractor; isolation; per-item HMAC). Schema-only and broken token overlap fail for all three. An English protocol or an equality recipe that does not name the field saturates. Silent same- σ overlap and vague hints are family-dependent, not a three-family restore. OpenAI silent 6/32 and vague-hint 13/32 include 24 and 15 empty completions stored as UNKNOWN. *Composer vague-hint distractor 1/32. JOIN was not evaluated at $n=32$. Wilson: 32/32 $\Rightarrow$ [0.89, 1.00], 0/32 $\Rightarrow$ [0.00, 0.11], 6/32 $\Rightarrow$ [0.09, 0.35], 31/32 $\Rightarrow$ [0.84, 0.99]. Silent same- σ OpenAI 6/32 is a JSON-binding cell, distinct from WIKI-H5. NS_SILENT ARM lines say “Match impossible”; those 0/32 rows are not a sealed-join collapse.

Arm (no written hop)	OpenAI	Composer 2.5	Grok 4.5
Schema list only	0/32	0/32	0/32
English protocol (no hop)	32/32	32/32	32/32
Same σ , silent	6/32	19/32	31/32
Same σ + equality recipe	32/32	32/32	32/32
JSONKEY: match broken	0/32	0/32	0/32
Sealed Q, two routes, recipe	31/32	25/32	32/32
Sealed Q, two routes, vague hints*	13/32	23/32	31/32

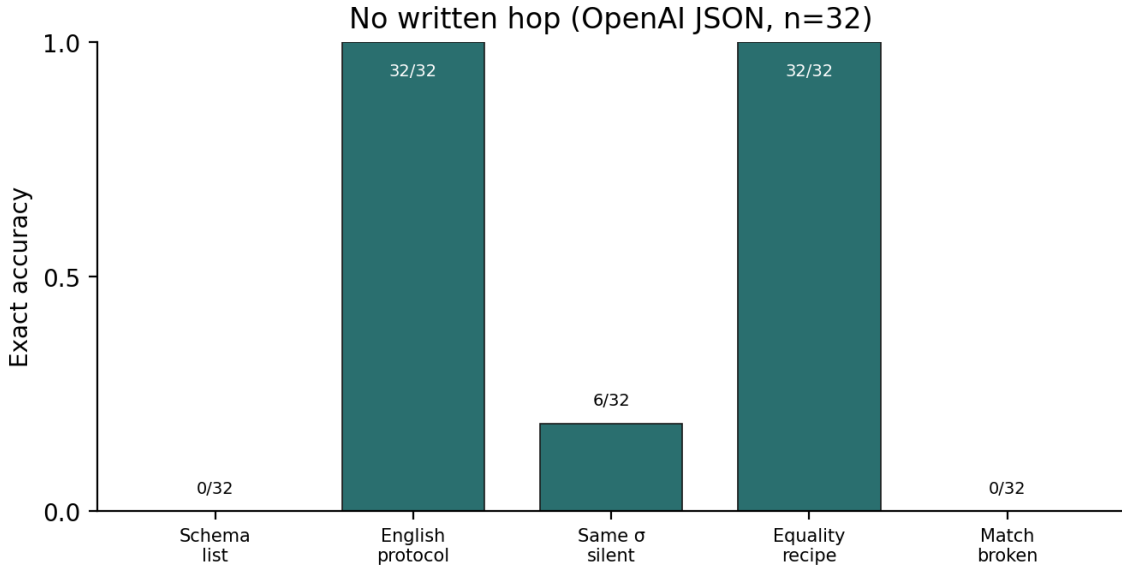


Figure 3: No-hop binding (Wikidata JSON, OpenAI $n=32$). Bars are exact match; the 6/32 silent bar is gold hits, not a decoy rate.

Figure 3 is the OpenAI row of Table 13. The Table 13 prompts are fixed English blocks (same wording for all three families). We use *equality* rather than similarity: exact identity of sealed tokens, not a continuous metric. Schema list only is a bilingual `KEY_SCHEMA` (English name = sealed JSON key) plus “Use the schema to read CONTEXT. No JOIN is given.” The English protocol adds that QUESTION and CONTEXT are encoded, that identical tokens are the same entity or field (equality, not decryption), and that schema names plus token equality walk CONTEXT; it also names the asked field as country rather than city. The same- σ equality recipe (Appendix J) does not name country or city in English. It is not the WIKI-H5 prompt. The suite uses the same 32 Wikidata people (per-item keys; seed 20260826–20260829). The OpenAI $n=6$ cell with an English question template and English JSON keys (1/6) is not plaintext Wiki-H5: the start token and all values are sealed, city is a nested distractor under `hq`, and the prompt forbids world knowledge and allows UNKNOWN. English keys alone do not restore the walk.

F Sealing middleware

The necessary step is binding a typed full name to one graph atom: spaced tokens otherwise miss V . The layer then seals entities, relations, and sensitive values. When the relation sequence is already specified, the join can be executed by a deterministic non-LLM router. If the hop is not compiled, the layer may attach one sealed join as an execution control (not a list of candidate paths). The model, when used, returns $\sigma(a)$; the layer unseals. Isolation on the CEO vault without start-entity binding is 0/6 vs. full layer 6/6 for Composer 2.5, OpenAI, and an unversioned Claude endpoint (Anthropic model id not stored; plaintext names in the packed prompt: 0). On complex 3-hop and real 2Wiki, Composer 2.5 without binding is 0/6 vs. layer 6/6; OpenAI layer 6/6 (without binding: complex 4/6, 2Wiki 0/6). Those isolation files also attach a join: that is the execution control, not the reason to call a model. The model never receives a retrieved path list and never answers in plaintext (Table 1). The 6/6 scores are this binding-and-join control, not the free WIKI-H5 6/32 cell.

G Packaging probes

These cells test serialization and application packaging. They are *not* the matched 2×2 , not a second discovery, and not required to restate the body claim. Sample sizes are $n=6$ –16 except a counterfactual

2Wiki sanity check and the public Spider identifier-in- q cell ($n=100$). We keep one paragraph per family so a reader who wants the packaging inventory can find it; a reader of WIKI-H5 can skip this appendix.

Tables / JSON / SQL / Excel / FinQA / WTQ. Same CEO join as markdown tables, nested JSON, or a value store: free sealed English 0/12, join plan 12/12 (Composer 2.5 and OpenAI). JSON keys without a legend: 0/6; a key legend yields 6/6 for OpenAI only; a join plan yields 6/6 for both. Token overlap vs. SQLCOL: break on public Spider is Table 14 (OpenAI; identifier-in- q slice $n=100$: 51/100 vs. 0/100). Excel NL already names both join predicates (batched $n=16$): Composer 0/16 (mid-hop error 16/16) vs. plan 16/16; OpenAI 16/16 on both. FinQA / WTQ (Chen et al., 2021; Pasupat and Liang, 2015) on shared sealed context: FinQA NL 0/8 vs. gold programs 8/8; WTQ NL 0/6 vs. LOOKUP DSL 6/6 (Composer). OpenAI matches FinQA and WTQ DSL; WTQ NL is 1/6. Not a leaderboard.

Counterfactual 2Wiki sanity check. Composer 2.5 plain NL 200/200 (Wikipedia leak 0); OpenAI batched 150/200 with overlap leak 50/200. We do not treat this as a standard 2Wiki evaluation. Spider: sealed-cell NL 4/12; model SQL executed locally 29/32; gold-SQL engine 191/200 (Appendix K).

Table 14: Spider identifier overlap vs. namespace break (OpenAI; Yu et al., 2018). Slice A $n=100$: EQ 51/100, NS 0/100. Not official exact-match. n.r. for Composer/Grok.

English q	Find the number of papers published by the institution “University of Pennsylvania”.
Sealed q (A)	Find the number of E41f626311a9f published by the institution “University of Pennsylvania”.
EQ	q token equals sqlite identifier $\rightarrow$ execute; first-cell gold on this item
NS	same q ; identifiers are <code>HMAC(SQLCOL:name)</code> $\rightarrow$ UNKNOWN
A (name in q)	$n=100$ (45 DBs). EQ 51/100; NS 0/100 (UNKNOWN 100/100)
B (no name in q)	$n=26$. EQ 18/26; NS 20/26 (English q ; legend can still bind)

Long context / rename. At $\sim 128k$ and $\sim 1.14M$ sealed tokens, free sealed NL remains 0/8; plans 8/8. Paired sealed plans under a second HMAC key: 12/12 with differing golds (Figure 4).

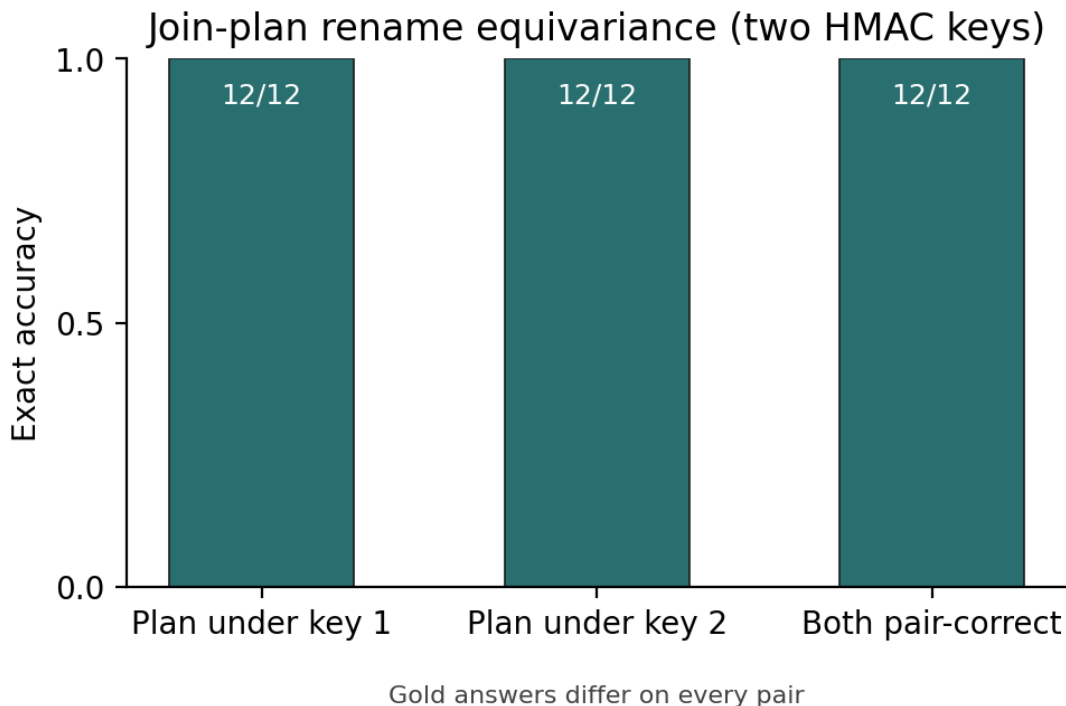


Figure 4: Sealed-plan rename equivariance ($n=12$): plans remain correct under a second HMAC key with a different gold on every pair.

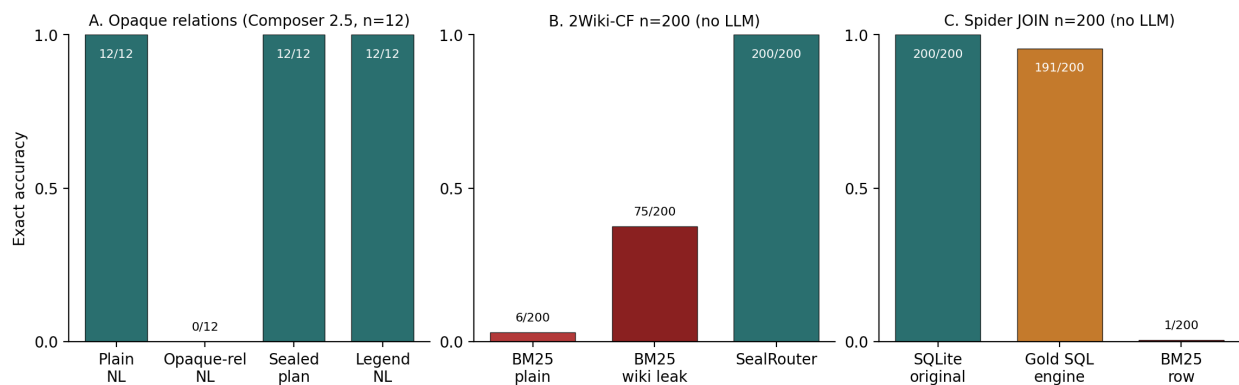


Figure 5: Non-LLM and packaging controls. Panel A is Composer $n=12$; panels B–C are frozen $n=200$ baselines. Left: plain NL 12/12, opaque-rel NL 0/12, plan/legend 12/12 (Composer 2.5, $n=12$). Middle: 2Wiki-CF BM25 6/200, Wikipedia-string leak 75/200, SealRouter 200/200. Right: Spider sqlite 200/200, gold-SQL engine 191/200, BM25 1/200. CEO/2Wiki opacity $\times$ prompt-form scores are in the surrounding prose.

H Extended scores

MetaQA-style 2-hop templates on the same public movie graph are a question-form control, not an evaluation on the official MetaQA 14K test split. Plaintext is 32/32 for all three families. Condition A (English question over opaque relations) is OpenAI 32/32, Composer 19/32, Grok 15/32. Condition B hashes those verbs except [start]: OpenAI 25/32, Composer 27/32, Grok 14/32 (plans 32/32, 31/32, 32/32). That $n=32$ template cell is not MOVIE-B. Packaging cells with $n \leq 16$ are Appendix G, not a second 2×2 .

I Sign-flip of the H5 interaction

Let $d_i = 1[\text{opaque correct}] - 1[\text{English correct}]$ and $c_i = d_i^{\text{two}} - d_i^{\text{unique}}$. Of 32 WIKI-H5 items, 24 have $c_i = -1$, one has $c_i = -2$, 7 have $c_i = 0$; $T = -26$. Under exchangeability of unique-path and two-path opacity effects, $p = 2^{-24} = 5.96 \times 10^{-8}$. That p coincides numerically with the two-path McNemar p (25 vs. 0; McNemar, 1947); the tests are not the same. Opacity is not a randomized within-item treatment. The body reports Wilson intervals and the paired counts.

J Equality recipe (JSON cell)

Verbatim protocol used in Table 13, *not* in WIKI-H5: “QUESTION and CONTEXT use the same encoding. Identical tokens are the same entity or the same field (equality, not English). START is a QUESTION token that also appears as an id value in CONTEXT. The asked field is a QUESTION token that also appears as a JSON key. Walk CONTEXT by token equality. If more than one reading is possible, UNKNOWN.”

The $n=32$ JSON cells are Table 13. Sample sizes $n \leq 16$ (vault, Excel, FinQA, WTQ, middleware) are packaging probes (Appendix G), not WIKI-H5.

K Symbolic execution controls

SealRouter executes a gold sealed path with no LLM. The gold-SQL engine executes gold Spider SQL locally. Figure 5 summarizes those controls.